

Problem D

Bánh Bò

Time limit: 15 seconds

Ever since the Earth got destroyed, Trillian has been missing some Earth delicacies. Today, she had the spaceship's food machine generate for her a Vietnamese delicacy she once enjoyed: bánh bò hấp (steamed chewy sponge cake).

Trillian has an unlimited number of *bánh bò hấp* pieces. Each piece of *bánh bò hấp* is either *red* or *white*. She wants to assemble rc pieces of *bánh bò hấp* into a grid with dimensions $r \times c$, where each cell contains a single piece of *bánh bò hấp*. Thus, there are exactly 2^{rc} distinct ways to assemble *bánh bò hấp* into an $r \times c$ grid, since we consider pieces of the same color to be identical.

We say an assembly of *bánh bò hấp* is *uniform* if all 6×7 subgrids have the same number of red pieces. Consequently, in a uniform *bánh bò hấp* assembly, all 6×7 subgrids have the same number of white pieces as well. Note that an $r \times c$ grid has $(r - 5)(c - 6)$ subgrids of dimensions 6×7 .

For example, Figure D.1 illustrates a uniform assembly of 7×8 pieces of *bánh bò hấp*, where shaded cells represent red *bánh bò hấp* pieces and unshaded cells represent white *bánh bò hấp* pieces. Figure D.2 shows that all four 6×7 subgrids have 6 red pieces and 36 white pieces.

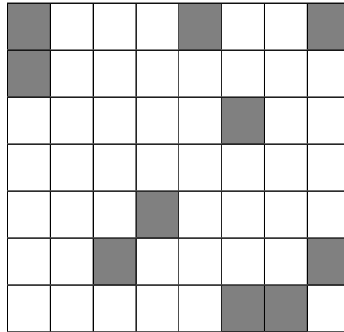


Figure D.1: An example of uniform *bánh bò hấp* assembly.

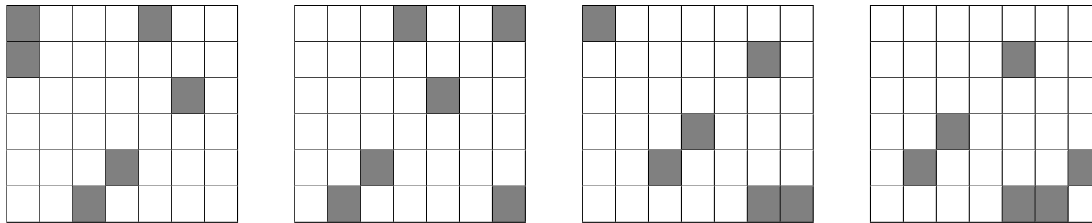


Figure D.2: All four 6×7 subgrids of the uniform *bánh bò hấp* assembly illustrated in Figure D.1.

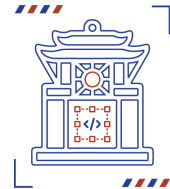
Given r and c , where r is a multiple of 6 and c is a multiple of 7, Trillian would like to calculate the number of possible uniform *bánh bò hấp* assemblies modulo 998 244 353.



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Input

Input consists of a single line containing two integers r and c ($6 \leq r \leq 66\,666$; r is a multiple of 6; $7 \leq c \leq 77\,777$; c is a multiple of 7).

Output

Output the number of possible uniform *bánh bò hấp* assemblies modulo 998 244 353.

Sample Input #1

6 7

Sample Output #1

780136139

Explanation for the sample input/output #1

The output is 2^{42} modulo 998 244 353.

Sample Input #2

12 14

Sample Output #2

22889737

Sample Input #3

12 42

Sample Output #3

96403614

Sample Input #4

42 14

Sample Output #4

94940316